HIGHER SCHOOL CERTIFICATE EXAMINATION

2023

Mathematics Advanced

Examiner: Sami El Hosri

General Instructions

- Reading Time – 10 minutes
- Working Time – 3 hours
- Write using black pen
- Calculators approved by NESA may be used
- A separate reference sheet is provided at the back of this paper.
- All necessary working should be shown in Questions 11 – 32.

Total marks – 100

Section I: Multiple Choice

Questions 1 – 10 10 marks

- Attempt all questions
- Allow about 15 minutes for this section.

Section II: Extended Response

Questions 11 – 32 90 marks

- Attempt all questions
- Allow about 2 hours 45 minutes for this section.

Section I
Questions 1 – 10 (10 marks)

Read each question and choose an answer A, B, C or D.
Allow about 15 minutes for this section.

Use the multiple-choice answer sheet for Questions 1-10.

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1 Which of the following relations is many to one?

A) $y = 2x + 3$

B) $y = 2$

C) $x = 2$

D) $y = x^3$

2 The function $f(x) = 2\sin 4x$ is to be dilated horizontally by a factor of $\frac{1}{4}$ and then translated horizontally $\frac{\pi}{8}$ units to the left.

What is the equation of the transformed function $h(x)$?

A) $h(x) = 2\sin 16\left(x + \frac{\pi}{8}\right)$

B) $h(x) = 2\sin\left(x + \frac{\pi}{8}\right)$

C) $h(x) = 2\sin\left(16x + \frac{\pi}{8}\right)$

D) $h(x) = 2\,\text{sins}\,16\left(x - \frac{\pi}{8}\right)$

3 The average mass of 6 bags of lollies is 0.4 kg.

What could be the mass of the heaviest bag?

A) 0.3 kg

B) 3 kg

C) 2.5 kg

D) 0.45 kg

4 ABC is an isosceles triangle such that AB = AC = 8 cm.

Given that the area of triangle ABC is 16 cm^2, find the possible values of angle ABC.

A) $30°, 150°$.

B) $15°, 70°$.

C) $15°, 75°$.

D) $30°, 120°$.

5 The functions $f(x)$ and $g(x)$ are defined for all real x and the function $g(x) = \left(f(x)\right)^3$.

Given that $f'(-1) = 9$, $g'(-1) = 432$ and $f(-1) > 0$,
what is the value of $f(-1)$?

A) 48

B) 4

C) 16

D) 12

6 The displacement of a particle moving along the x axis is given by

$x = t^3 + \ln\ (2t^3 + 1)$.

How many times is the particle at rest?

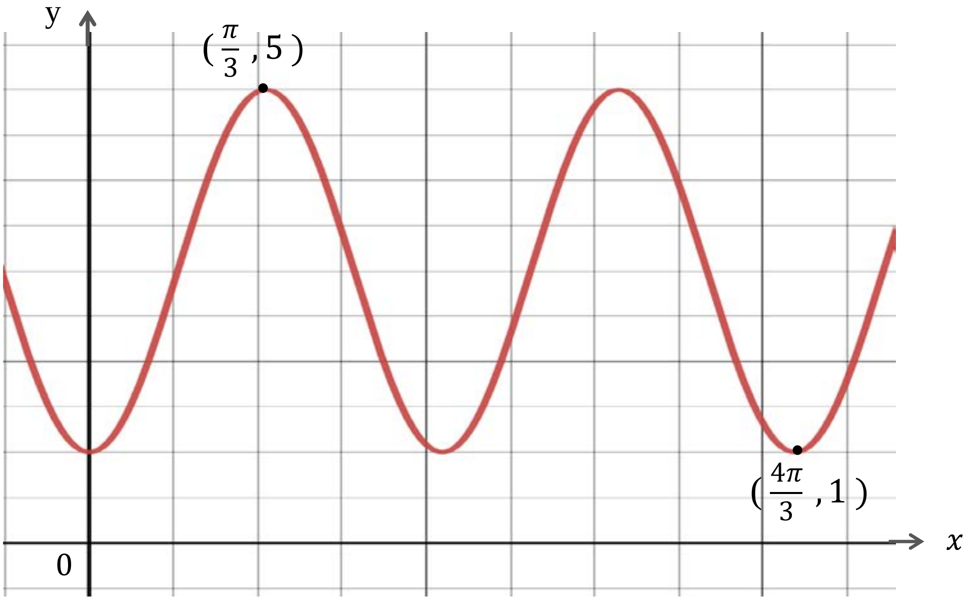
A) 3

B) 2

C) 1

D) 0

7 Which of the equations below is represented by the graph?



A) $y = 3 - 2\cos 2x$

B) $y = 3 - 2\cos 3x$

C) $y = 2 + 3\cos 3x$

D) $y = 3 + 2\cos 3x$

8 A company produces juice bottles. The volumes of the bottles are normally distributed with mean 400 mL.

The probability of selecting a bottle containing less than 395mL is a and the probability of selecting a bottle containing more than 410 mL is b.

Given that a bottle containing more than 395mL is selected at random, what is the probability it contains more than the mean and less than 410 mL?

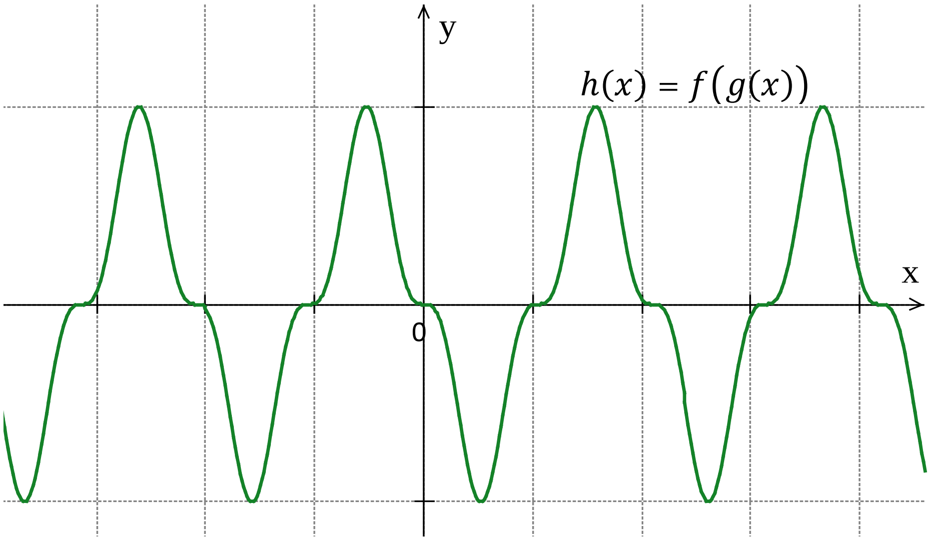
A) $\frac{1 - b - a}{1 - a}$

B) $\frac{1 - b}{2 - a}$

C) $\frac{1 - b}{1 - a}$

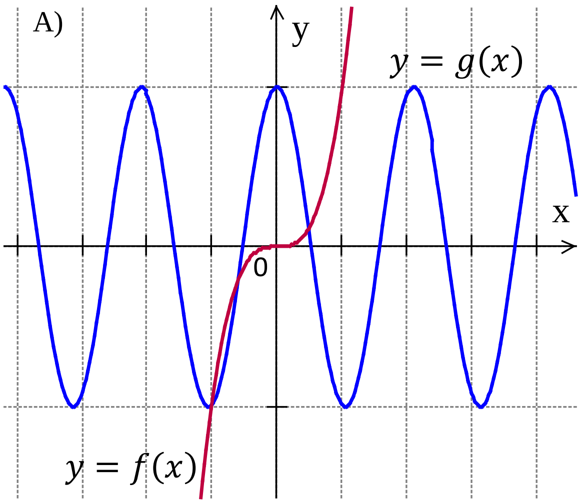
D) $\frac{1 - 2b}{2 - 2a}$

9. The diagram shows the graph of the composite function $h(x) = f\big(g(x)\big)$.

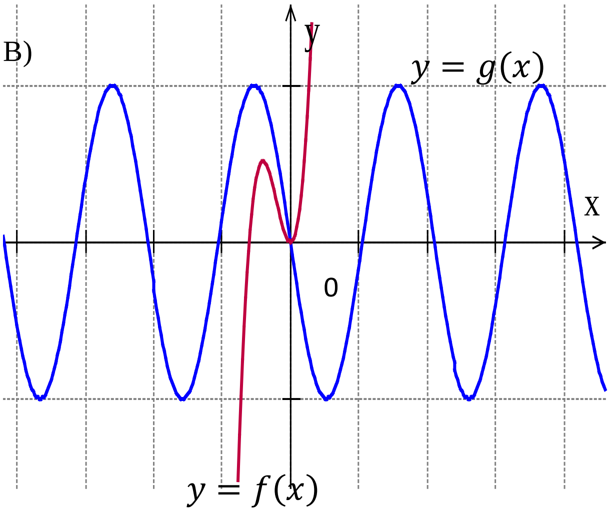


Which of the following is best option to represent the graphs of two functions $f(x)$ and $g(x)$?

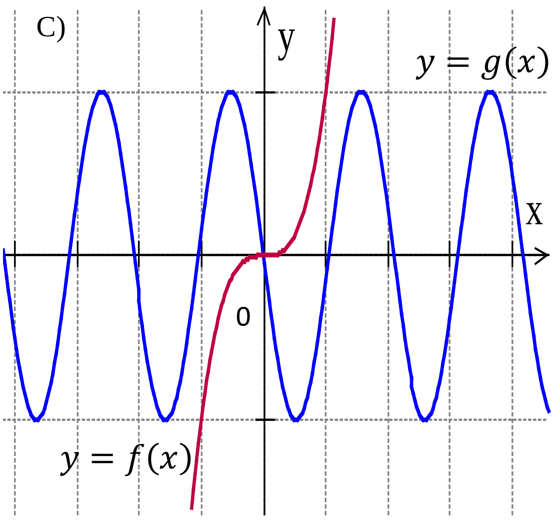
A)



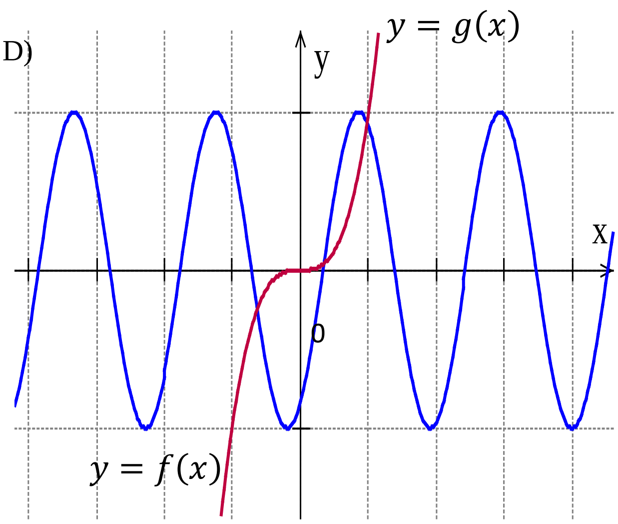
B)



C)



D)



10 The functions $y = f(x)$ and $y = g(x)$ are continuous for all x.

Given that $\int_0^a \left(f(x) + g\,(x)\right)^2 dx = b$ and $\int_0^a \left(f(x)\right)^2 + \left(g(x)\right)^2 dx = c$,

what is the value of $\int_0^a \left(1 + f(x)\, g\,(x)\right) dx$?

A) $\dfrac{2a + b - c}{2}$

B) $\dfrac{a + b - c}{2}$

C) $\dfrac{2a - b - c}{2}$

D) $\dfrac{2a + b + c}{2}$

ADVANCED MATHEMATICS

Section II

90 marks

Attempt Questions 11 – 32

Allow about 2 hours and 45 minutes for this section.

INSTRUCTIONS

- Answer the questions in the spaces provided. Sufficient spaces are provided for typical responses.
- Your responses should include relevant mathematical reasoning and/or calculations.
- Extra writing space is provided at the back of this booklet.
  If you use this space, clearly indicate which question you are answering.

Question 11 (2 marks)

Data was collected from 20 people on the number of messages that they made in the last month. The set of data collected is displayed in the stem and leaf plot. **2**

Messages

2	0
3	0 4
4	0 0 0 2 5 5 6
5	0 0 0 0 2 2 2
6	0 0
7	0

Is 20 an outlier for this set of data? Justify your answer using suitable calculations.

Question 12 (2 marks)

The table shows the compounded value of $1 at different interest rates over different periods. **2**

Compounded values of $ 1

Number of periods	Interest rate per period			
	1%	2%	3%	4%
4	1.0406	1.0824	1.1255	1.1699
6	1.0615	1.1262	1.1941	1.2653
8	1.0829	1.1717	1.2668	1.3686
10	1.1046	1.2190	1.3439	1.4802

Cheryl wants to buy a boat for $15 000. Based on the information provided, what is the minimum she must invest at 4% p.a. over 4 years, compounding half yearly, to achieve her goal?

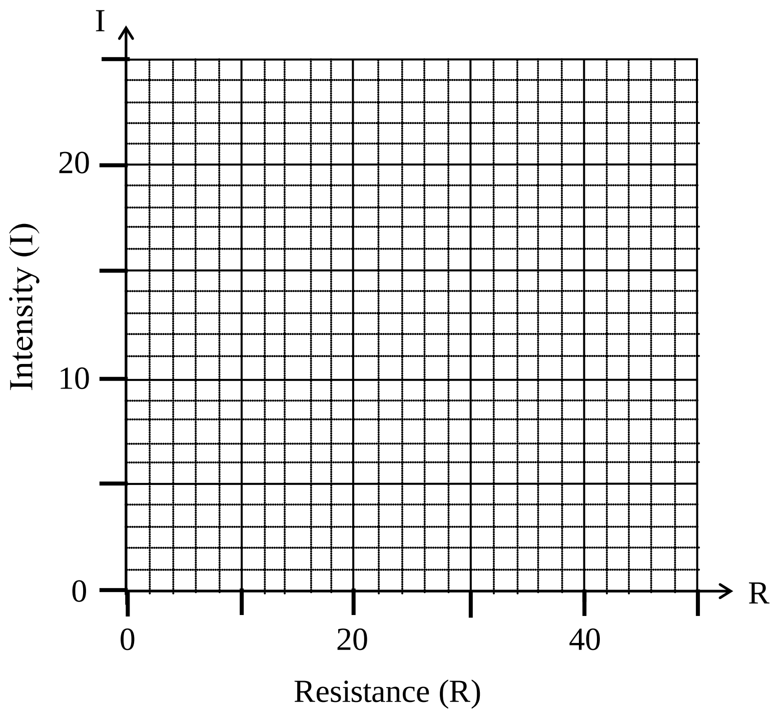
Question 13 (5 marks)

The intensity I of a current varies inversely with the resistance R in a circuit.
The intensity $R = 10$ when $I = 24$.

a) Find the equation of I in terms of R. **2**

b) Complete the table of values then graph the relationship between the intensity of a current and the resistance, from $R = 10$ to $R = 50$. 3

R	10	30	40	50
I				



Question 14 (3 marks)

Use the trapezoidal rule, with 5 function values, to find the area between the curve $y = x\sin^3 x$, the x axis, and the lines $x = 0$ and $x = \pi$. 3

Give your answer correct to 3 decimal places.

Question 15 (2 marks)

The derivative function of a curve is $f'(x) = 6e^{3x} - 4\sin 2x$. **2**

The curve passes through the origin.

Find the equation of the curve $y = f(x)$.

Question 16 (4 marks)

In a game, there are three different spinning wheels labelled A, B and C.

Each of the three wheels is divided into small sectors, each of equal size.

In each sector a player can get either "Win Holiday "or "Win car".

The information about the spinning wheels is as shown in the table.

Daniel is to play the game and he is to spin a wheel at random.

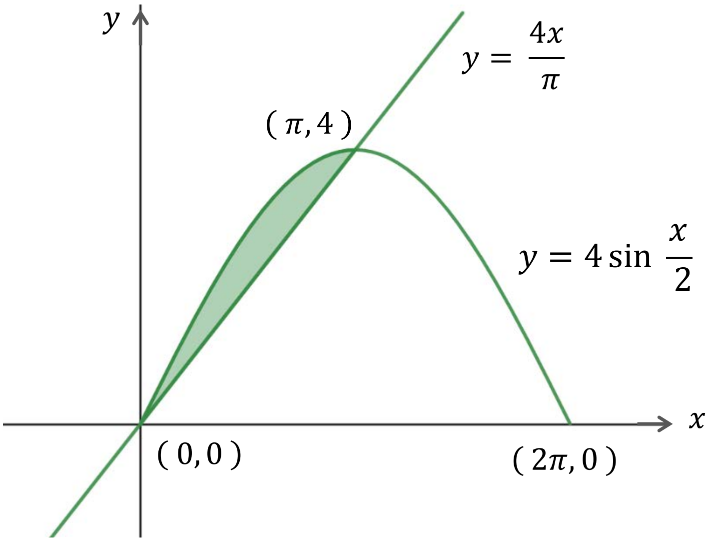
	Spinning Wheels		
	A	B	C
Win Holiday	6	2	8
Win Car	3	6	2
Total	9	8	10

a) What is the probability that Daniel selects wheel A and Win Car? **2**

b) Given that Daniel wins a car, what is the probability that he selected wheel C? **2**

Question 17 (3 marks)

The curve $y = 4\sin\frac{x}{2}$, $0 \le x \le 2\pi$, meets the line $y = \frac{4x}{\pi}$ at the points $(0, 0)$ and $(\pi, 4)$, as shown in the diagram. **3**



Find the exact value of the shaded area.

Question 18 (5 marks)

A new technology director recorded the profits of her company for the first 20 weeks of her job.

The company's profits in the first week was $8000. In each subsequent week the

profit was $2 400 more than the previous week.

a) Calculate that the profit the company makes in the 20th week. **2**

b) Calculate the least number of weeks the company needs to make a total profit exceeding $1 000 000? **3**

QUESTION 19 (3 marks)

ABC is an acute angled triangle such that AB = 12 cm and AC = 10 cm. **3**

The area of the triangle is 30 cm ^{2}.

Find the perimeter of the triangle correct to 2 decimal places.

QUESTION 20 (4 marks)

The table below shows the future value of annuity with a contribution of \$1 at the end of each period.

Period	0.75%	1.00%	1.50%	2.00%	2.50%	3.00%
1	1.00000	1.00000	1.00000	1.00000	1.00000	1.00000
2	2.00750	2.01000	2.01500	2.02000	2.02500	2.03000
3	3.02256	3.03010	3.04523	3.06040	3.07563	3.09090
4	4.04523	4.06040	4.09090	4.12161	4.15252	4.18363
5	5.07556	5.10101	5.15227	5.20404	5.25633	5.30914
6	6.11363	6.15202	6.22955	6.30812	6.38774	6.46841
7	7.15948	7.21354	7.32299	7.43428	7.54743	7.66246
8	8.21318	8.28567	8.43284	8.58297	8.73612	8.89234

Kevin opened an account and deposited \$15 000 at the end of each year for 6 years.

At the end of the 7^{th} year, he deposited \$20 000 in the same account and continued to deposit \$20 000 at end of each year until the end of the 10^{th} year.

He made no more deposits after the end of the 10^{th} year.
The money in his account was earning interest at 2.5% per annum compounded annually.

How much money will Kevin have in his account at the end of 15 years? **4**

Question 21 (3 marks)

The masses of salt bags, produced by a factory, are normally distributed with a mean of 800 g and a standard deviation of 9 g. **3**

It is known from statistical tables that for this distribution approximately 9.1% of these bags contain more than 812 g.

What is the approximate percentage of these bags between 782 g and 788 g?

Question 22 (4 marks)

Consider the function of a curve is $f(x) = x^3 - 12x - 6$, where $-3 \leq x \leq 5$. **4**

Find the global minimum and maximum values of $f(x)$.

Question 23 (3 marks)

Data was collected to study the relationship between the number of alcohol drinks per day. and the reduction in life expectancy, in years, for many drinkers from various regions in a country.

a) The correlation coefficient of this data is 0.9264. **1**

What does this indicate about the relationship between the number of alcohol drinks per day and the reduction in life expectancy for the drinkers?

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b) Using the values in the tables below find that the equation of the least-squares line of best fit for the reduction in life expectancy for these drinkers. **2**

	Mean	Standard deviation
Number of alcohol drinks per day	$\bar{x} = 6$	$\sigma_x = 2.43$
Reduction in the life expectancy (years)	$\bar{y} = 2.42$	$\sigma_y = 1.82$

The equation of the least squares line of best fit is y = gradient $\times\, x + y$ intercept	$\text{gradient} = r \times \dfrac{\text{standard deviation of } y \text{ scores}}{\text{standard deviation of } x \text{ scores}}$ $y\text{-intercept} = \overline{y} - (\text{gradient} \times \overline{x})$

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Question 24 (5 marks)

Consider the curve $f(x) = x - \sin x$, where $0 \leq x \ \leq 4\pi$.

a) Explain why the curve passes through the origin. **1**

b) Sketch the graph of showing the coordinates of any stationary points and the coordinates of the points of inflection. **4**

Question 25 (5 marks)

At the start of this year the number of a rare type of birds N in a national park was predicted to follow the formula $N = 5\,000 - 120\log_2(\,32t + 64\,)$,
where t is time in years from the start of this year.

a) What was the number of birds at the start of this year? **1**

b) At what rate was the number of birds decreasing at the start of the year? **2**

c) After how many years will the number of birds will be 3 920 ? **2**

Question 26 (4 marks)

P $(-4, -5)$ is a point on the function $f(x)$. The point Q is the image of point P on the function $g(x) = 2f\left(\frac{1}{2}x - 4\right)$.

a) What are the coordinates of Q? **2**

b) $N(a, b)$ is a point on the function $f(x)$. **2**

$M(6, -4)$ is the image of point N on the function $g(x)$.

What are the coordinates of N ?

Question 27 (4 marks)

A particle initially at rest at the origin starts moving along the *x* axis.

Its acceleration $a\ ms^{-2}$ at a time t seconds is given by $a = \dfrac{1}{2t+3}$.

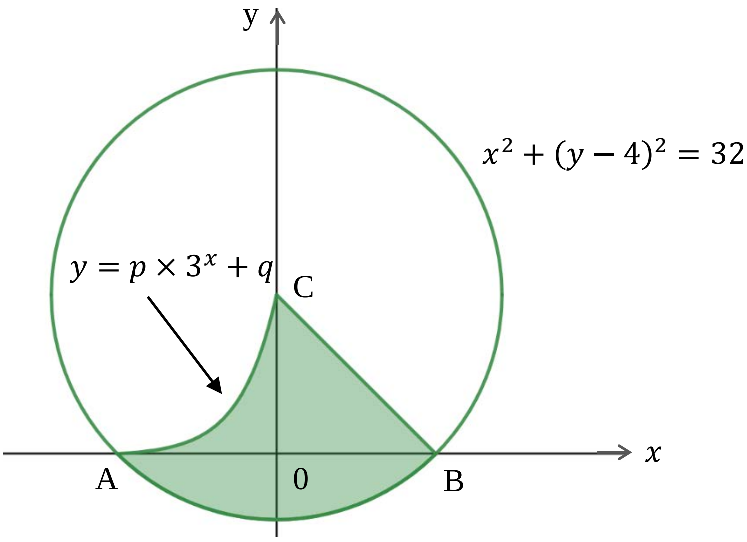
a) Show that the velocity $v\ ms^{-1}$ of the particle at time t is $v = \dfrac{1}{2} ln\left(\dfrac{2t+3}{3}\right)$. **2**

b) Find the time taken by the particle reach a velocity of $ln\,3\ \ ms^{-1}$. **2**

Question 28 (7 marks)

The shaded region in the diagram is bounded by the parts of the line $y = 4 - x$, the circle $x^2 + (y-4)^2 = 32$ and the curve $y = p \times 3^x + q$, where p and q are constants.

The circle has centre at C and intersects the x axis at A and B.



a) Find the coordinates of A, B and C. **2**

b) Show that $p = \frac{81}{20}$ and $q = -\frac{1}{20}$ **2**

Question 28 continues on page 22

c) Find the exact area of the shaded region. 3

End of Question 28

Question 29 (4 marks)

A continuous random variable, X, has the following probability density function.

$$f(x) = \begin{cases} k\sqrt{2}\,cos\,kx & \text{for } 0 \le x \le \pi. \\ 0 & \text{for all other values of } x. \end{cases}$$

a) Given that $0 < k < \frac{1}{2}$, calculate the value of k. **2**

b) Given that P $(0 \le x \le p) = \frac{1}{\sqrt{2}}$, find the value of p. **2**

Question 30 (4 marks)

Kate borrows \$600 000 to be repaid at a reducible rate of 0.35% per month.
Her monthly payment is \$3 500.

Let A_n be the amount in dollars she still owes at the end of n months.

$A_n = 200\,000\,(\,5 - 2 \times 1.0035^{\,n}\,)$ (DO NOT prove this)

a) Find the total amount of interest paid in the first year. **2**

b) Calculate the least number of months Kate needs to drop her loan under \$300 000. **2**

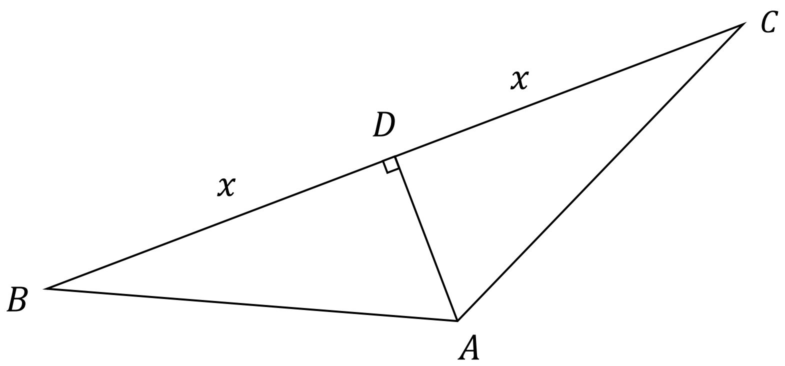
Question 31 (7 marks)

ABC is an isosceles triangle with AB = AC.

D is the midpoint of BC and AD is perpendicular to BC.

The perimeter of the triangle is k metres, where k is a constant.

Let $DC = BD = x$ metres.



a) Show the area of triangle ABC is $A = x\sqrt{\frac{k^2}{4} - kx}$. **3**

Question 31 continues on page 26

b) Given that the maximum area of triangle ABC occurs when $x = 30\ m$, find the value of k the perimeter of the triangle. **4**

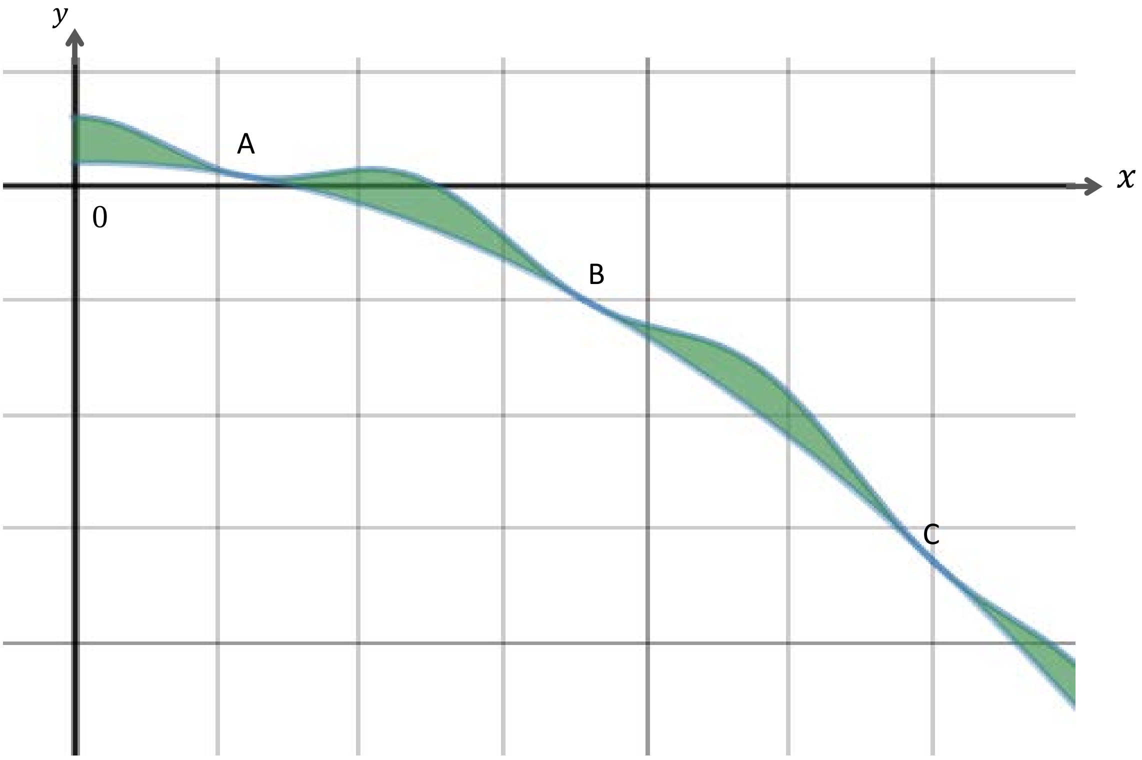
End of Question 31

Question 32 (7 marks)

Consider the functions $f(x) = \cos\frac{\pi x}{6} - \frac{x^2}{50} + \frac{x}{50} + 2$ and $g(x) = -\frac{x^2}{50} + \frac{x}{50} + 1$

in the domain $x \geq 0$. The x coordinates of the points of intersections of these two functions form the terms of an arithmetic sequence.

The points A , B and C shown in diagram below are the first points of intersections and their x coordinates are the first three terms of the arithmetic sequence.



a) Find the x coordinates of the points A, B and C. **2**

Question 32 continues on page 28

b) The sum of all the x coordinates of all the points of intersections of $f(x)$ and $g(x)$ in the domain $0 \leq x \leq p$ is 34 656. **2**

Show that the smallest value that p can be is 906.

c) Find the sum of all the shaded areas bounded by $f(x)$ and $g(x)$ in the domain $0 \leq x \leq 906$ **3**

END OF PAPER

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NSW Education Standards Authority

2023 HIGHER SCHOOL CERTIFICATE EXAMINATION

Mathematics Advanced
Mathematics Extension 1
Mathematics Extension 2

REFERENCE SHEET

Measurement

Length

$$l = \frac{\theta}{360} \times 2\pi r$$

Area

$$A = \frac{\theta}{360} \times \pi r^2$$

$$A = \frac{h}{2}(a+b)$$

Surface area

$$A = 2\pi r^2 + 2\pi rh$$

$$A = 4\pi r^2$$

Volume

$$V = \frac{1}{3}Ah$$

$$V = \frac{4}{3}\pi r^3$$

Functions

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

For $ax^3 + bx^2 + cx + d = 0$:

$$\alpha + \beta + \gamma = -\frac{b}{a}$$

$$\alpha\beta + \alpha\gamma + \beta\gamma = \frac{c}{a}$$

and $\alpha\beta\gamma = -\frac{d}{a}$

Relations

$$(x-h)^2 + (y-k)^2 = r^2$$

Financial Mathematics

$$A = P(1+r)^n$$

Sequences and series

$$T_n = a + (n-1)d$$

$$S_n = \frac{n}{2}[2a + (n-1)d] = \frac{n}{2}(a+l)$$

$$T_n = ar^{n-1}$$

$$S_n = \frac{a(1-r^n)}{1-r} = \frac{a(r^n-1)}{r-1},\ r \neq 1$$

$$S = \frac{a}{1-r},\ |r| < 1$$

Logarithmic and Exponential Functions

$$\log_a a^x = x = a^{\log_a x}$$

$$\log_a x = \frac{\log_b x}{\log_b a}$$

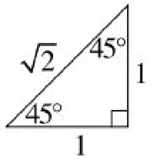
$$a^x = e^{x \ln a}$$

Trigonometric Functions

$$\sin A = \frac{\text{opp}}{\text{hyp}}, \quad \cos A = \frac{\text{adj}}{\text{hyp}}, \quad \tan A = \frac{\text{opp}}{\text{adj}}$$

$$A = \frac{1}{2}ab\sin C$$

$$\frac{a}{\sin A} = \frac{b}{\sin B} = \frac{c}{\sin C}$$

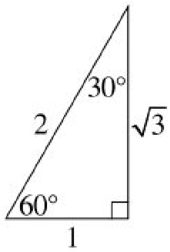


$$c^2 = a^2 + b^2 - 2ab\cos C$$

$$\cos C = \frac{a^2 + b^2 - c^2}{2ab}$$

$$l = r\theta$$

$$A = \frac{1}{2}r^2\theta$$



Trigonometric identities

$$\sec A = \frac{1}{\cos A}, \ \cos A \neq 0$$

$$\operatorname{cosec} A = \frac{1}{\sin A}, \ \sin A \neq 0$$

$$\cot A = \frac{\cos A}{\sin A}, \ \sin A \neq 0$$

$$\cos^2 x + \sin^2 x = 1$$

Compound angles

$$\sin(A + B) = \sin A\cos B + \cos A\sin B$$

$$\cos(A + B) = \cos A\cos B - \sin A\sin B$$

$$\tan(A + B) = \frac{\tan A + \tan B}{1 - \tan A\tan B}$$

If $t = \tan\frac{A}{2}$ then $\sin A = \frac{2t}{1+t^2}$

$$\cos A = \frac{1-t^2}{1+t^2}$$

$$\tan A = \frac{2t}{1-t^2}$$

$$\cos A\cos B = \frac{1}{2}\left[\cos(A - B) + \cos(A + B)\right]$$

$$\sin A\sin B = \frac{1}{2}\left[\cos(A - B) - \cos(A + B)\right]$$

$$\sin A\cos B = \frac{1}{2}\left[\sin(A + B) + \sin(A - B)\right]$$

$$\cos A\sin B = \frac{1}{2}\left[\sin(A + B) - \sin(A - B)\right]$$

$$\sin^2 nx = \frac{1}{2}(1 - \cos 2nx)$$

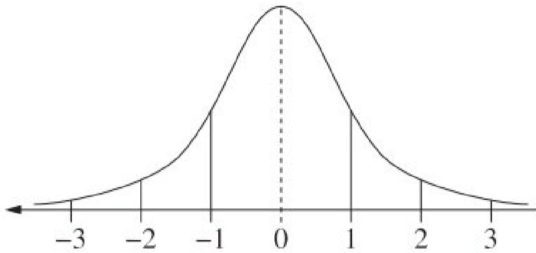
$$\cos^2 nx = \frac{1}{2}(1 + \cos 2nx)$$

Statistical Analysis

$$z = \frac{x - \mu}{\sigma}$$

An outlier is a score
less than $Q_1 - 1.5 \times$
or
more than $Q_3 + 1.5$

Normal distribution



- approximately 68% of scores have z-scores between −1 and 1
- approximately 95% of scores have z-scores between −2 and 2
- approximately 99.7% of scores have z-scores between −3 and 3

$$E(X) = \mu$$

$$\text{Var}(X) = E\left[(X - \mu)^2\right] = E\left(X^2\right) - \mu^2$$

Probability

$$P(A \cap B) = P(A)P(B)$$

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

$$P(A|B) = \frac{P(A \cap B)}{P(B)}, \ P(B) \neq 0$$

Continuous random variables

$$P(X \le x) = \int_a^x f(x)\,dx$$

$$P(a < X < b) = \int_a^b f(x)\,dx$$

Binomial distribution

$$P(X = r) = {}^nC_r p^r(1 - p)^{n-r}$$

$$X \sim \text{Bin}(n, p)$$

$$\Rightarrow \ P(X = x)$$

$$= \binom{n}{x}p^x(1 - p)^{n-x}, \ x = 0, 1, \ldots, n$$

$$E(X) = np$$

$$\text{Var}(X) = np(1 - p)$$

Differential Calculus

Function	Derivative
$y = f(x)^n$	$\frac{dy}{dx} = nf'(x)[f(x)]^{n-1}$
$y = uv$	$\frac{dy}{dx} = u\frac{dv}{dx} + v\frac{du}{dx}$
$y = g(u)$ where $u = f(x)$	$\frac{dy}{dx} = \frac{dy}{du} \times \frac{du}{dx}$
$y = \frac{u}{v}$	$\frac{dy}{dx} = \frac{v\frac{du}{dx} - u\frac{dv}{dx}}{v^2}$
$y = \sin f(x)$	$\frac{dy}{dx} = f'(x)\cos f(x)$
$y = \cos f(x)$	$\frac{dy}{dx} = -f'(x)\sin f(x)$
$y = \tan f(x)$	$\frac{dy}{dx} = f'(x)\sec^2 f(x)$
$y = e^{f(x)}$	$\frac{dy}{dx} = f'(x)e^{f(x)}$
$y = \ln f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{f(x)}$
$y = a^{f(x)}$	$\frac{dy}{dx} = (\ln a)f'(x)a^{f(x)}$
$y = \log_a f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{(\ln a)f(x)}$
$y = \sin^{-1} f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{\sqrt{1-[f(x)]^2}}$
$y = \cos^{-1} f(x)$	$\frac{dy}{dx} = -\frac{f'(x)}{\sqrt{1-[f(x)]^2}}$
$y = \tan^{-1} f(x)$	$\frac{dy}{dx} = \frac{f'(x)}{1+[f(x)]^2}$

Integral Calculus

$$\int f'(x)[f(x)]^n dx = \frac{1}{n+1}[f(x)]^{n+1} + c$$

where $n \neq -1$

$$\int f'(x)\sin f(x)\,dx = -\cos f(x) + c$$

$$\int f'(x)\cos f(x)\,dx = \sin f(x) + c$$

$$\int f'(x)\sec^2 f(x)\,dx = \tan f(x) + c$$

$$\int f'(x)e^{f(x)}dx = e^{f(x)} + c$$

$$\int \frac{f'(x)}{f(x)}dx = \ln|f(x)| + c$$

$$\int f'(x)a^{f(x)}dx = \frac{a^{f(x)}}{\ln a} + c$$

$$\int \frac{f'(x)}{\sqrt{a^2-[f(x)]^2}}dx = \sin^{-1}\frac{f(x)}{a} + c$$

$$\int \frac{f'(x)}{a^2+[f(x)]^2}dx = \frac{1}{a}\tan^{-1}\frac{f(x)}{a} + c$$

$$\int u\frac{dv}{dx}dx = uv - \int v\frac{du}{dx}dx$$

$$\int_a^b f(x)\,dx$$

$$\approx \frac{b-a}{2n}\left\{f(a) + f(b) + 2\left[f(x_1) + \cdots + f(x_{n-1})\right]\right\}$$

where $a = x_0$ and $b = x_n$

Combinatorics

$$^nP_r = \frac{n!}{(n-r)!}$$

$$\binom{n}{r} = {}^nC_r = \frac{n!}{r!\,(n-r)!}$$

$$(x+a)^n = x^n + \binom{n}{1}x^{n-1}a + \cdots + \binom{n}{r}x^{n-r}a^r + \cdots + a^n$$

Vectors

$$\left|\underset{\sim}{u}\right| = \left|x\underset{\sim}{i} + y\underset{\sim}{j}\right| = \sqrt{x^2+y^2}$$

$$\underset{\sim}{u}\cdot\underset{\sim}{v} = \left|\underset{\sim}{u}\right|\left|\underset{\sim}{v}\right|\cos\theta = x_1x_2 + y_1y_2,$$

where $\underset{\sim}{u} = x_1\underset{\sim}{i} + y_1\underset{\sim}{j}$

and $\underset{\sim}{v} = x_2\underset{\sim}{i} + y_2\underset{\sim}{j}$

$$\underset{\sim}{r} = \underset{\sim}{a} + \lambda\underset{\sim}{b}$$

Complex Numbers

$$\begin{aligned} z = a + ib &= r(\cos\theta + i\sin\theta) \\ &= re^{i\theta} \end{aligned}$$

$$\begin{aligned} \left[r(\cos\theta + i\sin\theta)\right]^n &= r^n(\cos n\theta + i\sin n\theta) \\ &= r^ne^{in\theta} \end{aligned}$$

Mechanics

$$\frac{d^2x}{dt^2} = \frac{dv}{dt} = v\frac{dv}{dx} = \frac{d}{dx}\left(\frac{1}{2}v^2\right)$$

$$x = a\cos(nt+\alpha) + c$$

$$x = a\sin(nt+\alpha) + c$$

$$\ddot{x} = -n^2(x-c)$$